

DARSHANEASE

Temple Darshan Booking Platform — MERN Stack
Project Documentation

Repository: github.com/bumireddyakshaya-source/DARSHANEASE

Demo Video: youtu.be/oLQMEwgLoiI

1. Introduction

Project Title: DARSHANEASE — Temple Darshan Booking Platform

Team Members:

- MUNASA VANAJA
- BHUMIREDDY AKSHAYA

2. Project Overview

Purpose

DARSHANEASE is a full-stack MERN application that lets users explore temples, view live darshan slot availability, book tickets, manage bookings, and make donations, with secure JWT authentication and role-based access control for three user types: USER, ADMIN, and ORGANIZER.

Features

- Browse and search temples by name or city
- View live darshan slot availability by date, with general and VIP slot types
- Book darshan tickets with live seat-count and total price calculation
- View and cancel bookings from a "My Bookings" page
- Make donations and view donation history
- Secure JWT-based authentication with role-based access (USER / ADMIN / ORGANIZER)
- Admin panel to add/update/delete temples and darshan slots
- Responsive Bootstrap 5 UI with a custom saffron/maroon temple theme

3. Architecture

Frontend

Built with React and Bootstrap 5. Uses React Router v6 for navigation with protected routes (PrivateRoute, RoleRoute) for logged-in-only and admin-only pages. A React Context (AuthContext) manages global authentication state without prop drilling. A pre-configured Axios instance handles API calls, automatically attaching the JWT and logging the user out on token expiry. React Toastify provides success/error notifications.

Backend

Built with Node.js and Express.js, structured into config, models, middleware, controllers, and routes layers. Middleware handles JWT authentication and role-based authorization (authorize ("ADMIN", "ORGANIZER")). A centralized error-handling middleware normalizes Mongoose validation, cast, and duplicate-key errors into clean JSON responses. CORS is locked to the configured CLIENT_URL.

Database

MongoDB is used with Mongoose models for: User, Temple, Darshan Slot, Booking, and Donation. A seed script (seed/seedTemples.js) populates sample data: 6 temples (Tirupati, Meenakshi, Golden Temple, Kashi Vishwanath, Siddhivinayak, Vaishno Devi), 5 days of darshan slots per temple (general + VIP), and a default admin account.

4. Setup Instructions

Prerequisites

Tool	Version	Check with
Node.js	v16 or higher	node -v
npm	Comes with Node	npm -v
MongoDB	Community Server (local) or free MongoDB Atlas cluster	Mongodb --version
Git (optional)	Any recent version	git --version
Postman (optional)	Any recent version	For testing APIs

Installation — Quick Setup

A one-command setup script is included.

macOS / Linux:

```
cd DARSHANEASE
chmod +x setup.sh
./setup.sh
```

Windows:

Double-click setup.bat, or run it from Command Prompt: setup.bat

The script installs backend and frontend dependencies, creates .env files from the examples, seeds the database with sample temples/slots/admin, and starts both servers. MongoDB must be installed and running (or an Atlas URI configured) before the script reaches the seeding step.

Installation — Manual Setup

1. Create a Folder with frontend and backend You will get a DARSHANEASE/ folder containing backend/ and frontend/.

2. Backend setup:

```
cd DARSHANEASE/backend
npm install
cp .env.example .env
```

Update backend/.env with:

```
MONGO_URI=mongodb://127.0.0.1:27017/DARSHANEASE
JWT_SECRET=DARSHANEASE_super_secret_key_change_this
JWT_EXPIRES_IN=7d
PORT=5000
CLIENT_URL=http://localhost:3000
```

Start MongoDB (skip if using Atlas), then seed and start the backend:

```
mongod
npm run seed
npm run dev # or: npm start
```

3. Frontend setup (in a new terminal):

```
cd DARSHANEASE/frontend
npm install
```

```
cp .env.example .env
npm start
```

Confirm frontend/.env points to the backend:

```
REACT_APP_API_URL=http://localhost:5000/api
```

5. Folder Structure

Client (frontend/)

```
frontend/
├── public/index.html
├── src/
│   ├── api/axios.js           # Pre-configured axios instance
│   │   (JWT interceptor)
│   ├── context/AuthContext.js
│   ├── components/           # Navbar, Footer, TempleCard,
│   │   SlotModal, PrivateRoute
│   ├── pages/                # Home, Login, Register,
│   │   Temples, TempleDetail,
│   │   |                     # MyBookings, Donations,
│   │   AdminDashboard, AdminAddTemple
│   └── App.js / App.css / index.js / index.css
```

Server (backend/)

```
backend/
├── config/db.js
├── models/                   # User, Temple, DarshanSlot, Booking,
├── middleware/               # auth (JWT), role (RBAC),
├── errorHandler
├── controllers/              # Business logic for each module
├── routes/                   # API route definitions
├── seed/seedTemples.js
└── server.js                 # App entry point
```

6. Running the Application

- Backend: npm start (or npm run dev for auto-reload) in the backend directory — runs at <http://localhost:5000>
- Frontend: npm start in the frontend directory — opens automatically at <http://localhost:3000>

7. API Documentation

Base URL: <http://localhost:5000/api>

Method	Endpoint	Access	Description
POST	/auth/register	Public	Register new user
POST	/auth/login	Public	Login, returns JWT
GET	/auth/me	Private	Get logged-in profile
GET	/temples	Public	List temples (search, city, pagination)
GET	/temples/:id	Public	Temple details
POST	/temples	ADMIN/ORGANIZER	Create temple
PUT	/temples/:id	ADMIN/ORGANIZER	Update temple
DELETE	/temples/:id	ADMIN	Delete temple
GET	/slots/temple/:templeId	Public	Slots for a temple (optional ?date=)
POST	/slots	ADMIN/ORGANIZER	Create darshan slot
PUT	/slots/:id	ADMIN/ORGANIZER	Update slot
DELETE	/slots/:id	ADMIN/ORGANIZER	Delete slot
POST	/bookings	Private	Book a darshan slot
GET	/bookings/my	Private	My bookings
GET	/bookings	ADMIN/ORGANIZER	All bookings
PUT	/bookings/:id/cancel	Owner/ADMIN	Cancel a booking
POST	/donations	Private	Make a donation
GET	/donations/my	Private	My donations
GET	/donations	ADMIN/ORGANIZER	All donations

Private routes require an Authorization: Bearer <token> header, using the token returned from login/register. Import the endpoints above into Postman for manual testing.

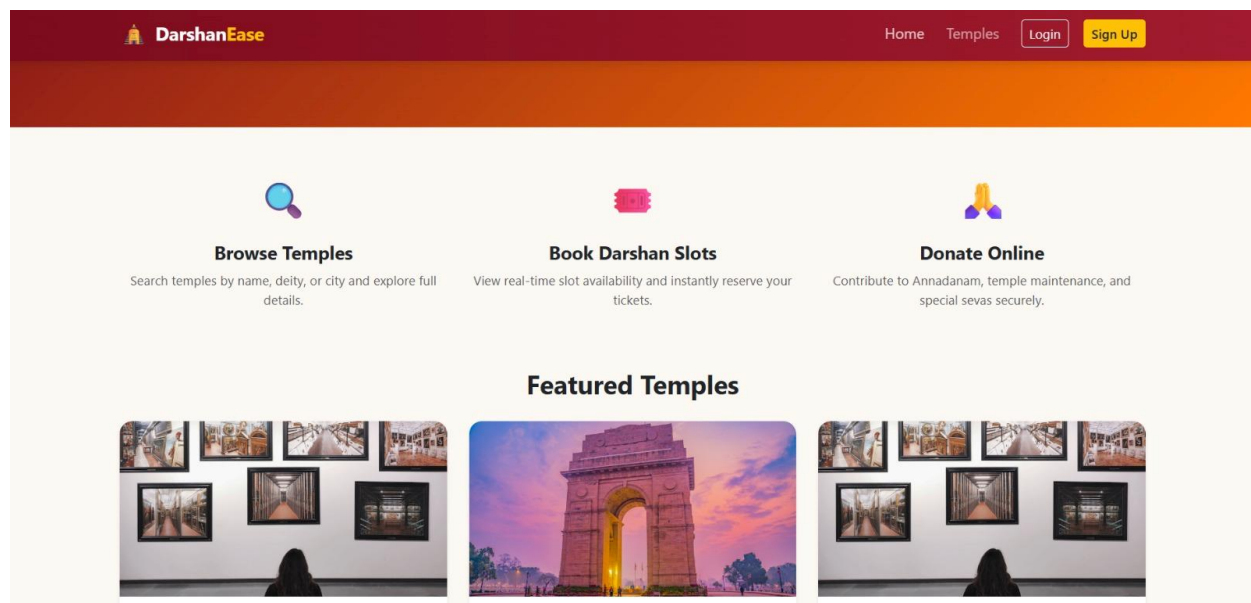
8. Authentication

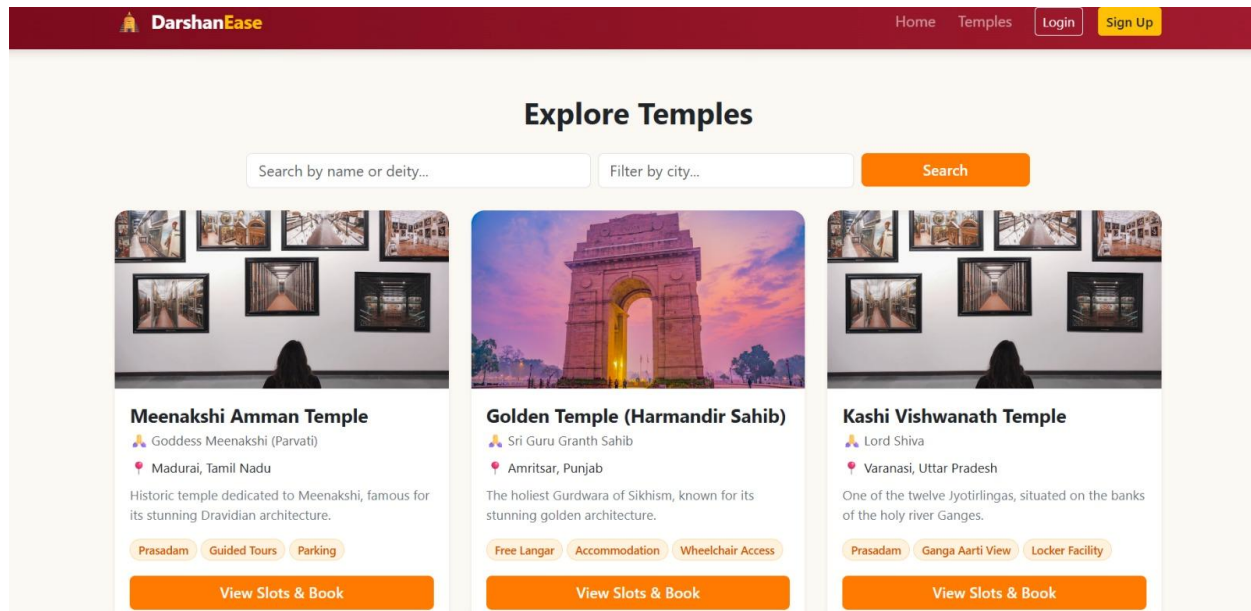
- Passwords are hashed with bcrypt.js before storage.
- A JWT is issued on login/register and verified on every protected route.
- Role-based middleware (authorize ("ADMIN", "ORGANIZER")) restricts sensitive routes such as temple/slot management and viewing all bookings/donations.
- The frontend Axios instance auto-attaches the JWT to requests and auto-logs-out the user on token expiry.
- CORS is locked to the configured CLIENT_URL for additional protection.

Default admin login (after seeding): admin@DARSHANEASE.com / Admin@123.
Regular users self-register from the Signup page, defaulting to the USER role.
ORGANIZER accounts are created by registering normally and then updating the role directly in MongoDB (or via an Admin-driven promotion flow in production).

9. User Interface

Key pages include Home, Temples (browse/search), Temple Detail (with slot booking modal), My Bookings, Donations, and, for admins, an Admin Dashboard and Add Temple page. The UI uses a fully responsive Bootstrap 5 layout with a custom saffron/maroon temple theme.





10. Testing

Manual testing is performed via a functional checklist covering the core user and admin flows:

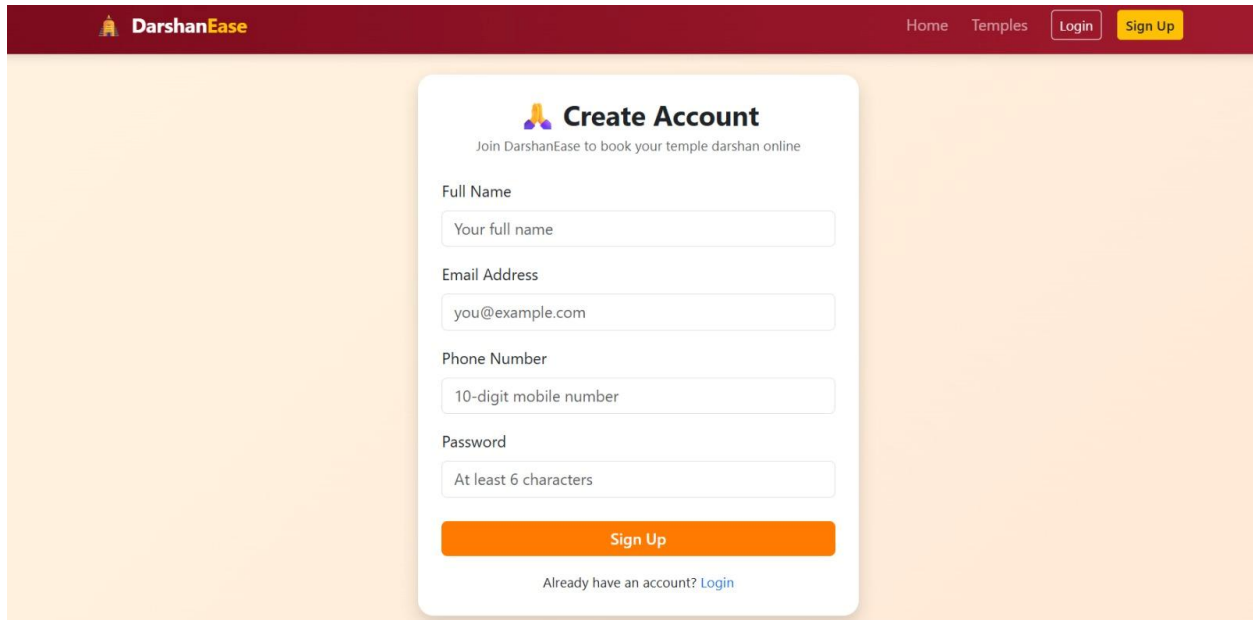
- Register a new user → login → check profile via /auth/me
- Browse temples, search by name/city
- Open a temple, filter slots by date, book a slot, check ticket count decreases
- View booking in My Bookings, cancel it, confirm slot capacity is released
- Make a donation, confirm it appears in donation history
- Login as admin, add a temple, add a slot, verify it appears publicly
- Try accessing /admin as a normal USER — should redirect to Home
- Try calling admin-only API routes without a token — should return 401/403

API endpoints are additionally tested manually via Postman.

11. Screenshots or Demo

Demo Video:

<https://drive.google.com/file/d/1kn1UdsJbHcLKgc9p31pNqtfEeRklnMrC/view?usp=sharing>



The screenshot shows the 'Create Account' form on the DarshanEase website. The header is dark red with the logo and navigation links: Home, Temples, Login, and Sign Up. The form is centered on a light orange background. It includes fields for Full Name, Email Address, Phone Number, and Password, each with a placeholder text. A 'Sign Up' button is at the bottom, and a link to 'Login' is provided for existing users.

Create Account
Join DarshanEase to book your temple darshan online

Full Name
Your full name

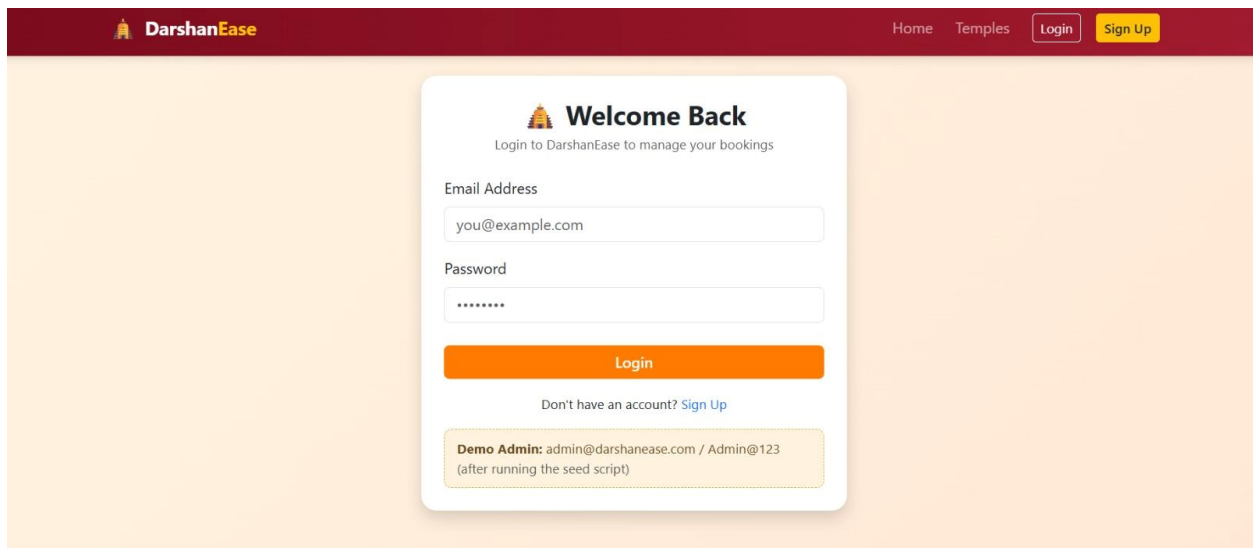
Email Address
you@example.com

Phone Number
10-digit mobile number

Password
At least 6 characters

Sign Up

Already have an account? [Login](#)



The screenshot shows the 'Welcome Back' login form on the DarshanEase website. The header is dark red with the logo and navigation links: Home, Temples, Login, and Sign Up. The form is centered on a light orange background. It includes fields for Email Address and Password, each with a placeholder text. A 'Login' button is at the bottom, and a link to 'Sign Up' is provided for new users. A demo admin login information box is also present.

Welcome Back
Login to DarshanEase to manage your bookings

Email Address
you@example.com

Password

Login

Don't have an account? [Sign Up](#)

Demo Admin: admin@darshanease.com / Admin@123
(after running the seed script)

DarshanEase

[Home](#) [Temples](#) [Login](#) [Sign Up](#)

Explore Temples



Meenakshi Amman Temple

 Goddess Meenakshi (Parvati)

 Madurai, Tamil Nadu

Historic temple dedicated to Meenakshi, famous for its stunning Dravidian architecture.

[Prasadam](#) [Guided Tours](#) [Parking](#)

[View Slots & Book](#)

DarshanEase

[Home](#) [Temples](#) [Login](#) [Sign Up](#)

Available Darshan Slots



Tue, 14 Jul VIP

 06:00 - 07:00

 ₹300/person

50 tickets available

[Book Now](#)

Tue, 14 Jul GENERAL

 09:00 - 10:00

 Free

100 tickets available

[Book Now](#)

Tue, 14 Jul GENERAL

 12:00 - 13:00

 Free

100 tickets available

[Book Now](#)

Tue, 14 Jul GENERAL

 16:00 - 17:00

 Free

100 tickets available

[Book Now](#)

Tue, 14 Jul GENERAL

 18:00 - 19:00

 Free

100 tickets available

[Book Now](#)

Wed, 15 Jul VIP

 06:00 - 07:00

 ₹300/person

50 tickets available

[Book Now](#)

Wed, 15 Jul GENERAL

 09:00 - 10:00

 Free

[Book Now](#)

Wed, 15 Jul GENERAL

 12:00 - 13:00

 Free

[Book Now](#)

Wed, 15 Jul GENERAL

 16:00 - 17:00

 Free

[Book Now](#)

12. Known Issues

Known Issues

- The application requires MongoDB to be running.
- Temple data must be seeded before using the application for the first time.
- Online payment functionality is not implemented.
- Email and SMS notifications are not available.
- The project is currently designed for local deployment.

13. Future Enhancements

- Online payment gateway integration for bookings and donations
- Email/SMS notifications for booking confirmations and reminders
- QR-code based e-tickets for darshan slots
- Admin analytics dashboard for bookings and donations
- Self-service ORGANIZER role request/approval workflow